

Chapter 160 — Open Radical Prostatectomy

- **Most radical prostatectomies (RPs) in the United States and Europe are now robot-assisted**, despite no evidence of better cancer control, fewer technical complications, or better continence or potency.
- **Open surgery still matters**: its lessons transfer to robotic surgery, it serves systems without a robot, and it is the fallback on conversion.
- **Total surgical removal is the best cure for clinically localised prostate cancer**. RP is the only treatment shown in a randomised controlled trial (Bill-Axelson; Scandinavian Prostate Cancer Group-4, SPCG-4) to reduce metastasis and prostate cancer death.
- **Better understanding of periprostatic anatomy** → less bleeding, fewer technical complications, better continence and potency.
- **Goals, in order of importance**: cancer control → minimise technical complications → continence → potency.
- **Open radical retropubic prostatectomy (ORRP) is among the hardest operations in urology**, demanding skill in both patient selection and technique.

Concept — *The order of goals settles every intraoperative trade-off: when nerve sparing and a clear margin conflict, the margin wins.*

RADICAL RETROPUBIC PROSTATECTOMY: SURGICAL ANATOMY

Venous and Arterial Anatomy

Venous drainage

- **Prostatic veins drain into the Santorini plexus**. Knowing it precisely is what gives a bloodless field at the membranous urethra and apex.
- **Deep dorsal vein**: leaves the penis beneath Buck's fascia between the corpora → pierces the urogenital diaphragm → three branches: **superficial branch + right and left lateral venous plexuses**.
- **Superficial branch**: central vein between the puboprostatic ligaments over the bladder neck and prostate; seen early; communicates over the bladder and into the endopelvic fascia; lies **outside the anterior prostatic fascia**.
- **Common trunk and lateral plexuses**: **hidden beneath the prostatic and endopelvic fascia**. The plexuses run posterolaterally and communicate freely with the pudendal, obturator and vesical plexuses; near the puboprostatic ligaments small branches pierce the sidewall muscle to the internal pudendal vein.
- **Lateral plexus → inferior vesical vein → internal iliac vein**. Because the veins anastomose freely, **any laceration of these friable veins can bleed heavily**.
- The **common trunk lies directly over the urethra** — the site where the dorsal vein is transected.

Arterial supply

- **The inferior vesical artery supplies the prostate**. After small branches to the seminal vesicle, bladder base and prostate, it ends in two groups:

TERMINAL PROSTATIC ARTERIES: URETHRAL VS CAPSULAR GROUPS

Feature	Urethral group	Capsular group
Entry or course	Enters at the posterolateral vesicoprostatic junction	Along the pelvic sidewall in the lateral pelvic fascia , posterolateral to the prostate

Feature	Urethral group	Capsular group
Territory	Vesical neck and periurethral gland	Outer prostate (ventral and dorsal branches); ends as a small cluster to the pelvic floor
Surgical significance	Divided at the bladder neck (5- and 7-o'clock pedicles)	Wrapped in a nerve network → macroscopic landmark for the microscopic cavernous nerves (the NVB)

- **Corpora cavernosa are supplied mainly by the internal pudendal artery. Accessory pudendal arteries** from the obturator, inferior or superior vesical arteries run over the lower bladder and anterolateral prostate, so RP divides them → reduced penile inflow, **worst in older men with borderline penile flow.**

Pelvic Plexus

- **Autonomic supply to pelvic organs and external genitalia.** Formed by **parasympathetic** preganglionic visceral efferents from the **sacral centre (S2–S4)** + **sympathetic** fibres from the **thoracolumbar centre** via the hypogastric nerve.
- **Location:** retroperitoneal beside the rectum, **5–11 cm from the anal verge**; a fenestrated rectangular sagittal plate with its **midpoint at the seminal vesicle tip.**
- **Inferior vesical vessels perforate the plexus** → ligating the "lateral pedicle" in its midportion cuts both vessels and the nerves to prostate, urethra and corpora.
- **Visceral branches:** bladder, ureter, seminal vesicles, prostate, rectum, membranous urethra, corpora. **Somatic motor axons** also pass through to the levator ani, coccygeus and striated urethral muscle.
- **All these nerves stay outside the capsule:** prostatic nerves run outside the capsule and Denonvilliers fascia until they pierce it; branches to the membranous urethra and corpora run in the lateral pelvic fascia, dorsolateral to the prostate, between prostate and rectum.
- **Neurovascular bundle (NVB), "NVB of Walsh":** capsular vessels + cavernous nerves. The nerves are invisible; the vessels mark where they are.
- **Cavernous branches join the capsular vessels in a spray-like pattern, forming the NVB 20–30 mm distal to the prostatovesical junction,** in the lateral pelvic fascia **between prostatic and levator fascia.**
- **At the apex, branches to the cavernous bodies and sphincter spray anteriorly and posteriorly,** with wide variation, some leaving the vessels to run anteriorly — the anatomical basis of high anterior release.
- Beyond the urogenital diaphragm the nerves pass **behind the dorsal penile artery and nerve** into the corpora.

Striated Urethral Sphincter

- **Not a horizontal "sandwich" but a vertical tubular sheath** (Oelrich) around the membranous urethra.
- **Why:** in utero it runs unbroken from bladder to perineal membrane; the prostate, growing from the urethra, invades and thins it, and part atrophies.
- **Adult:** horseshoe-shaped fibres at the apex form a tube, fusing in the posterior midline near the apex. The prostate does not sit "like an apple on a shelf" on a flat diaphragm; the **sphincter attaches broadly over the apical prostatic fascia** — which is why apical dissection and urethral reconstruction decide continence.
- At the apex the **NVBs lie posterolateral to the circumferential sphincter.**

PASSIVE VS ACTIVE CONTINENCE

Feature	Passive continence	Active continence
Muscle	Striated urethral sphincter	Levator ani (levator urethrae, pubourethralis) around apex and membranous urethra, inserting into the perineal body

Feature	Passive continence	Active continence
Fibre type	Fatigue-resistant, slow-twitch	Voluntary contraction

Innervation **Mainly pudendal nerve for both**, plus somatic motor fibres via the pelvic plexus

- **Postoperative pelvic floor exercises contract the levator ani**; because the sphincter shares its nerve supply, it is trained too.

Pelvic Fascia

THE THREE FASCIAL INVESTMENTS OF THE PROSTATE

Layer	Position	Key features	Surgical rule
Denonvilliers fascia	Between anterior rectum and prostate; extends up over the posterior seminal vesicles	Thickest at base and seminal vesicles, thinning caudally to end at the sphincter. Its layers cannot be separated microscopically	Excise completely for an adequate margin
Prostatic fascia ("capsule")	Continuous with parenchyma anteriorly and anterolaterally	Carries dorsal vein tributaries and Santorini plexus anteriorly	Must stay on the prostate in nerve sparing
Levator fascia	Covers pelvic muscles; fuses laterally with prostatic fascia as the lateral pelvic fascia ; parts posterolaterally to follow the perirectal muscles	Vessels and autonomic nerves reach the prostate between levator and prostatic fascia	Incised or released to reach the NVB

- **Perineal vs retropubic**: perineal RP reflects the lateral and anterior fasciae off the prostate, sparing the dorsal vein and Santorini plexus → **less bleeding**; retropubic RP approaches from outside the fasciae → **dorsal vein complex (DVC) must be ligated and lateral pelvic fascia divided**.

Concept — *The anatomy is one sandwich: nerves, arteries and veins run between levator and prostatic fascia. Leaving the prostatic fascia on the gland and Denonvilliers fascia on its back is what lets nerve sparing and clear margins coexist.*

SURGICAL TECHNIQUE

Preoperative Preparation

- **Assessment**: focused systems review; full medical, surgical and anaesthetic history; all medicines, prescribed or not; examination; routine blood tests.
- **Rare life-threatening risks** — myocardial infarction, stroke, arrhythmia, pulmonary embolism, haemorrhage, anaesthetic reaction. Identify at-risk men to reduce risk or offer alternatives.
- **Factors that make surgery harder but do not contraindicate it**: previous abdominal or pelvic surgery or radiotherapy, transurethral surgery, extensive biopsies, significant inflammatory bowel disease, hernia mesh, and prostate size.
- **Inguinal hernia**: **about 15% have one on proper preoperative examination; up to 20% develop one after ORRP**. Most were silent before, so ORRP seems to unmask them → **examine standing with Valsalva and repair preperitoneally at ORRP**. Standing Valsalva ultrasonography finds even more; with this approach, symptomatic hernias afterwards are rare.
- **Defer surgery 6 weeks after needle biopsy and 12 weeks after transurethral resection of the prostate (TURP)**, so adhesions and haematoma settle and planes return to near normal — crucial for NVB preservation and avoiding rectal injury.

- **No autologous donation:** transfusion is rare and donated units go unused unless anaemic, so donation only deepens postoperative anaemia. **Preoperative erythropoiesis-stimulating agents** raise haematocrit and reduce anaemia; higher discharge haematocrit speeds recovery.
- **Prophylaxis when the urinary tract is entered:** first- or second-generation cephalosporin, or aminoglycoside + metronidazole or clindamycin.
- **Antiplatelet and anticoagulant plans** are made with the internist or cardiologist. **Drug-eluting coronary stents** pit **stent thrombosis (usually catastrophic)** against bleeding; **low-dose aspirin** often gives enough cover without extra bleeding at ORRP.
- **Bowel preparation:** clear fluids the day before → half a bottle of magnesium citrate that evening → enema that morning; same-day admission.
- **Pretreatment magnetic resonance imaging (MRI)** can guide how much NVB and membranous urethra to preserve.

Special Instruments

- **Fibreoptic headlight: essential** — much of the operation is beneath the pubis, where the view is poor.
- **Balfour retractor** with narrow and wide malleable blades: helpful for lymphadenectomy, **required for RP** (cranial and posterior retraction of peritoneum and bladder).
- Optional: coagulating forceps, small, fine and standard right-angled clamps, Metzenbaum and Jamison scissors, **2.5–4.5× loupes** (help the apical dissection).

Anaesthesia, Incision, and Lymphadenectomy

- **General endotracheal anaesthesia. Relative hypotension (systolic ≤ 100 mm Hg) and crystalloid ≤ 1500 mL until the prostate is out.**
- **Supine;** flex the table in obese men to lengthen the umbilicus–pubis distance.
- **16-Fr silicone (Silastic) Foley,** 20 mL balloon, closed drainage — the small calibre eases mucosal suturing. A right-handed surgeon stands on the patient's left.
- **Incision:** extraperitoneal lower midline, pubis towards umbilicus → anterior fascia to pubis → recti separated → transversalis fascia opened → **space of Retzius** → peritoneum lifted off external iliacs to the common iliac bifurcation, **vas left intact** → Balfour placed.
- **Leave the tissue over the external iliac artery:** it carries the leg lymphatics; dividing it risks **leg oedema and lymphocele**.
- **Exposure:** narrow malleable blade under the vas lifts the peritoneum; deep Deaver retracts the bladder medially.
- **Pelvic lymph node dissection (PLND) precedes RP for intermediate- and high-risk disease;** omitted for well-differentiated or low-volume favourable intermediate-risk disease (Gleason Grade Group 1–2).
- **PLND sequence** (dominant-tumour side first): divide adventitia over the external iliac vein, **sparing the arterial lymphatics** → under the vein to the sidewall → down to the femoral canal, ligating lymphatics (**node of Cloquet need not be removed**) → up to the common iliac bifurcation, clearing the external iliac–hypogastric angle → **obturator nodes, protecting the obturator nerve;** obturator vessels skeletonised, ligated only if bleeding → down to the pelvic floor, exposing hypogastric veins → repeat contralaterally.
- **Extended dissection** removes more nodes than a limited one → **better staging and possible therapeutic benefit.**

Exposure

- **Malleable blade lifts the peritoneum** and gently displaces the bladder back, exposing the anterior prostate.

Incision in the Endopelvic Fascia

- Clear fibroadipose tissue → expose pelvic fascia, puboprostatic ligaments, superficial dorsal vein.
- **Open the endopelvic fascia at its sidewall reflection, well away from bladder and prostate**, where it is transparent over the levator ani.
- **Too medial → lateral Santorini plexus → persistent venous bleeding**; the prostatic arteries and pelvic plexus branches lie beneath it.
- Extend anteromedially towards the puboprostatic ligaments → the lateral prostate becomes palpable.
- **Pudendal perforators through the pelvic muscle → clip, don't coagulate**: the pudendal artery and nerve lie just deep, along the pubic ramus.

Division of the Puboprostatic Ligaments

- Tease fibrofatty off the superficial vein and ligaments → free the vein from the medial ligament edge → coagulate and divide it → sponge stick pushes the prostate back → divide each ligament with scissors.
- Go **deep enough to expose the apex-anterior DVC junction**, where the DVC will be divided.
- **Spare the pubourethral component**: it anchors the striated sphincter anteriorly to the pubis.

Preservation of Accessory Pudendal Arteries

- **Arterial insufficiency contributes to erectile dysfunction after nerve sparing**; accessory pudendals over the anterolateral prostate are one cause.
- **Prevalence depends on the method**: about 70% in cadavers, about 7% on selective angiography; **large visible vessels in about 4%**.
- **Releasing them can bleed briskly**, forcing prompt DVC ligation → **complete the opposite side first** when a prominent vessel is unilateral.
- **Sequence**: divide endopelvic fascia lateral to the vessel and the puboprostatic ligament (vessel runs beneath it) → vessel loop → free the artery sharply with a right-angled clamp, lifting the anterior prostatic fascia → divide venous tributaries (bleeding controlled by running 4-0 absorbable) → continue caudally past the planned DVC division.

Ligation and Division of the Dorsal Vein Complex

- **Aim**: divide the DVC with minimal bleeding, **without injuring the sphincter or entering the anterior apex**.

CONTROL OF THE DORSAL VEIN COMPLEX: STAPLER VS SUTURE LIGATION

Feature	Endovascular stapler (current preference)	Suture ligation (traditional)
Sequence	Babcock bunches endopelvic fascial edges → Endo GIA (vascular load) about 4 cm over the DVC → move the Foley in and out; fire only if it moves freely (urethra not caught) → residual vein oversewn with 3-0 Monocryl	3-0 Monocryl through the DVC just distal to the apex → needle reversed → through pubic symphysis perichondrium → figure-of-eight horizontal mattress, tied, kept long → Babcock bunches fascia → figure-of-eight 2-0 on anterior prostate near bladder neck (proximal back-bleeding, worse with incompetent valves)
Blood loss	About 200–400 mL less	Controls most venous bleeding
Why it helps	Less traction → less stretch injury to NVBs and sphincter; three staple rows each side seal both ends; apical staple line covers anterior apical tumours	Flat, unbunched surface easier to divide; recreates puboprostatic support of the sphincter; fixes the DVC anteriorly , exposing the anterior apical plane

- **Stance for all apical sutures**: face the head of the table, needle driver against the pubis, perpendicular to the patient.

Division of the Dorsal Vein Complex

- Sponge stick presses the prostate down → Metzenbaum or No. 15 blade, **starting at the left edge**, where the apical junction shows best. The anteriorly fixed distal DVC opens the plane between prostate and striated muscle; loupes and resting the scissors on the anterior apex help find it.
- **Anterior apex = commonest site of positive margins**, because its surface is hard to see.
- **Caveat:** near the apex an **apical vessel tethers the NVB medially beneath the sphincter**, so cutting too close to the apex risks it → **divide the lateral sphincter edges obliquely, midway between apex and pelvic floor**.
- **Venous control must be absolute** so the rest is bloodless: the kept 3-0 Monocryl (5/8-circle) **oversews the distal sphincter-DVC edge with a loosely held driver, forming a hood over the anterior urethra; circumflex veins at 5 and 7 o'clock** → clips.
- **Proximal back-bleeding:** oversee the proximal DVC edges on the prostate in a V with the preplaced 2-0 — this also spreads tension evenly over the prostatic fascia, easing high release.

Division of the Urethra and Placement of Urethral Sutures

- **Standard nerve release** → **divide the urethra first; high anterior release** → **after nerve release**.
- **Sequence:** prostate pushed back → release lateral sphincter bands at their midpoint → **right-angled clamp around urethral smooth muscle close to the apex** (transect as near the apex as possible) → **divide anterior two-thirds**, sparing the catheter → place sutures → remove catheter, place 6-o'clock suture → divide posterior urethra.
- **The clamp separates posterior urethra from posterior sphincter** and, in a dry field, shows the sphincter-NVB relationship; the intact **posterior sphincter shields the NVBs**.
- **Six 3-0 Monocryl sutures** (5/8-circle tapered): text gives 1, 3, 5, 7, 9, 11 o'clock; operative figures show 12, 2, 5, 6, 7, 10 o'clock.
- **Mucosa and submucosa only — smooth-muscle bites delay continence.** The 16-Fr catheter makes the mucosa easy to see.
- Flimsy urethra → include the DVC in the first (anterior) suture for strength. The first sutures lift the mucosa, easing the rest.
- Placed outside → in; if inside → out is easier, a **French-eye needle** passes them from the bladder side at the end. Towels over the sutures prevent traction.

Posterior striated sphincter complex (*skeletal muscle + fibrous tissue*) — *precise division matters for four reasons:*

- (1) **apical margins**; (2) the **right plane on the anterior rectum, so all of Denonvilliers fascia is excised**; (3) **no blunt trauma to the NVBs** just behind it; (4) **continence**.
- **Sequence:** clamp under the **left edge midway between apex and urethra** (here the NVB has fallen back beneath the clamp) → divide → **right edge** → **centre** last.
- **Never divide it all from one side** — that endangers the contralateral NVB; a clamp too close to the apex endangers the NVB.

Identification and Preservation of the Nerve Bundle

- **Both NVBs can usually be preserved safely; excising both is rarely needed.** With modern technique and phosphodiesterase type 5 (PDE5) inhibitors, most healthy potent men **under 65** stay potent.
- **Extracapsular extension rarely exceeds 1–2 mm** — and that much tissue often remains even with the NVB preserved.

- **Nerves directly on the capsule enter the prostate; they are not cavernous nerves.** Cavernous branches lie behind the capsular vessels.

NVB DISSECTION PLANES COMPARED

Plane	Where the dissection runs	Margin risk	Verdict
Interfascial	Between levator and prostatic fascia; prostatic fascia left on the prostate	Low	Correct nerve-sparing plane
Intrafascial	Beneath prostatic fascia, on the parenchyma , to avoid capsular-vessel bleeding	High	Never used
Wide excision	NVB + lateral pelvic fascia taken with the specimen	Does not lower the overall margin rate	For induration, a fixed bundle or doubtful posterolateral margin

- **Excising the NVB does not prevent positive margins**, because most margins are elsewhere: **apex → posterior → posterolateral**; about a fifth are multiple. **Margin rates are the same with and without nerve sparing.**
- **Walsh series:** bilateral preservation 87%, one NVB excised 13%, **overall positive margins about 5%.** Penetration severe enough to need both excised implies probable distant spread and incurability; in practice one side usually proves free.
- **The decision is never final before surgery.** Inputs: **erectile status** — continence is the same after bilateral or unilateral sparing and whether or not erections return, so NVBs are spared less aggressively in men already impotent; **palpable apical lesion**; **high predicted penetration on Partin tables or MRI.**
- **Intraoperative triggers for wide excision on that side:** induration in the lateral pelvic fascia once the endopelvic fascia is open; **too little tissue over the posterolateral surface after removal** (NVB excised secondarily).

Identification of the Neurovascular Bundle

- **Touch is open surgery's advantage:** induration → excise; **bundle fixed to the prostate on gentle release → don't preserve.** Touch is dulled laparoscopically and absent robotically.
- **No upward traction;** roll the prostate side to side. With the catheter out the prostate is softer and the plane easier to find.

Standard Preservation of the Neurovascular Bundles

- **Starts after** urethral division, suture placement and posterior sphincter division.
- **Sequence:**
 - Babcock or sponge stick rolls the prostate over, so the **prostate is released from the bundle, not the bundle from the prostate** (less traction).
 - Right-angled clamp under the **superficial levator fascia from the bladder neck** (dividing its thick band there frees the prostate at once) to the apex; **loupes** confirm the prostatic fascia is intact and allow gentler dissection. Releasing laterally first makes the posterior apical release easier.
 - Locate the NVB by a **subtle posterolateral "groove"** → trace it to the apex, where the NVB turns caudolaterally to the urethra.
 - Find the medial border at the apex → develop the midline plane back to the rectum.
 - Release **from the rectal surface, apex → base**, spreading a right-angled clamp → clip and divide apical vessels → to the midpoint → to the seminal vesicle.
 - **Posterior arterial branch from the NVB over the seminal vesicle to the prostatic base** → ligate and divide each side → the NVBs fall back, untethered.

- **Apical kinking artefact:** with the prostate rolled across the midline the NVB looks kinked and **can be cut by mistake**; it straightens once apical vessels are divided.
- **NVB vessels: small clips parallel to the bundle; never thermal energy** (monopolar, bipolar, harmonic). The prostatic side of tiny vessels is simply cut with fine scissors.
- **Fixation not explained by vessels → excise the NVB.**
- **Why start at the apex:** branches are seen before they are cut; an **anteriorly placed bundle** could have its groove mistaken for the prostatorectal space; lateral veins joining Santorini plexus to NVB are easier to control.
- Working from the rectal side **keeps Denonvilliers and prostatic fascia on the gland** (only levator fragments come away) → iatrogenic margins in organ-confined disease become rare, and a mobile bundle can be felt and inspected → **partial excision with negative margin and preserved potency is possible even with extension near the NVB.**

High Anterior Release of the Neurovascular Bundles at the Apex

- **Purpose:** faster recovery of potency and continence by **less traction** on nerves to the cavernous bodies and sphincter, and by **not cutting small anterior branches.**
- **Risk: less apical soft tissue → possibly more positive margins**, so select men at low risk of extension near the NVB.

TSUZUKI CRITERIA: EXTRAPROSTATIC EXTENSION NEAR THE NVB (PER SIDE)

Criterion	Threshold
Prostate-specific antigen (PSA)	>10 ng/mL
Gleason score	>6
Average % of core involved, that side	>20%
% of cores with tumour, that side	>33%
Perineural invasion	Present
Digital rectal examination	Abnormal

- **≥2 criteria → >10% risk of extension near the NVB on that side → no high release there.**
- **Sequence:** after DVC ligation, before urethral incision, only with complete haemostasis → incise **levator fascia over the anterior apex** along the lateral DVC edge → extend distally along the lateral apex, **preserving the glistening white prostatic fascia** beneath the lateral veins → tributaries **clipped, never cauterised** → free levator fascia from the lateral "shoulders" → beyond the apex the NVB is seen lateral to the urethra → release it to just above the seminal vesicles → divide the DVC to the urethra between the two preplaced sutures.
- **Then:** urethra and sutures → posterior sphincter → if unilateral, a right-angled clamp under the contralateral apical levator fascia enters the same plane from inside.
- Excellent visualisation and magnification are essential to avoid entering the prostate.
- **Unilateral release recovers potency at 12 months as well as bilateral → now done on one side only, the pathologically more favourable one.** So the benefit comes from **more accurate, lower-traction preservation**, not from sparing anterior branches.

Wide Excision of the Neurovascular Bundles

- **Free the opposite NVB from the apex first**, to spare it traction injury.
- **Sequence:** identify NVB at apex → right-angled clamp **medial → lateral, directly on the anterior rectum** → divide **without ligation** to maximise soft tissue (clip distal end later if bleeding) → divide fascia on the lateral rectum, apex → base → end at the **seminal vesicle tip, where the NVB is ligated and divided.**

- **Passing the clamp lateral → medial makes rectal injury more likely.**
- NVB and lateral pelvic fascia come away under direct vision, generously covering the tumour.

Posterior Dissection and Division of the Lateral Pedicles

- Replace catheter → **light upward catheter traction** (possible now the NVBs are free) → divide the rectum–Denonvilliers attachment in the midline, **leaving all Denonvilliers layers over the seminal vesicles.**
- **Sponge stick on the catheter, never on the prostate** — displaced tissue can create a **false-positive margin.**
- **Divide the lateral pedicle on the lateral seminal vesicle** (NVB already clear) in **three layers: superficial → middle → deep; all at once risks cutting into the prostate.** Clip arterial bleeders — more tissue stays on the prostate and the NVBs are protected.
- Continue onto the anterolateral bladder–prostate junction → divide Denonvilliers fascia over the seminal vesicle tips; many surgeons now divide the vasa and free the seminal vesicles.

Division of the Bladder Neck and Excision of the Seminal Vesicles

- **Sequence:** incise **anterior bladder neck at the prostatovesical junction** → through mucosa → deflate balloon, clamp catheter ends together for traction → widen → divide **inferior vesical branches at 5 and 7 o'clock** → find the plane between **anterior seminal vesicles and posterior bladder** → divide posterior bladder neck hugging the seminal vesicles, **watching the ureteric orifices** → Allis on bladder neck → clip and divide vasa → free seminal vesicles → ligate tip arteries → divide residual Denonvilliers attachments → remove specimen.
- **Pelvic plexus lies on the lateral seminal vesicle** → dissect under direct vision, especially laterally, **staying on the seminal vesicle** and clipping small arteries.
- **Complete seminal vesicle excision is essential for cancer control.**
- **Inspect the specimen: doubtful posterolateral margin** → excise that NVB.
- **Near the NVB, clips not cautery:** a **rectovesical haematoma** inflames the NVBs and **delays sexual recovery** → a **small peritoneal window in the rectovesical cul-de-sac** lets any haematoma decompress.

Bladder Neck Reconstruction and Anastomosis

- **Tennis-racquet closure:** running or interrupted **2-0 absorbable, full thickness** (including mucosa prevents haematuria), **posterior midline forwards**, until the neck matches the urethra.
- **Ureteric orifices:** earlier indigo carmine helps; ureteric catheters usually unnecessary.
- **Mucosal eversion:** 4-0 absorbable advances mucosa over raw muscle → rosette → **mucosa-to-mucosa anastomosis. 6-o'clock suture left long** for urethral suture placement.
- **Intussusception** — buttressing sutures stop the neck pulling open as the bladder fills:
 - Remove nearby clips (they can fold into the neck → contracture) → **2-0 Maxon in the posterior bladder wall edges about 2 cm from the neck**, tied midline → slacken the malleable blade to show loose perivesical tissue → **second figure-of-eight 2-0 Maxon anteriorly, about 2 cm lateral to the neck each side**, tied.
 - **End point: "turtle sign"** — neck poking out beneath the anterior hood like a turtle's head. Saline fill → competent, little leak → empty with forceps through the neck.
 - With intussusception, most men are pad-free by 3 months and nearly all by 1 year.

- **Margin–continence trade-off:** this technique favours **ample apical margins** and may take more sphincter, so some men take longer to be pad-free. Preserving the puboprostatic ligaments and dissecting closer to the apex may spare sphincter but risks **inadequate apical margins**.
- **Anastomosis:** check haemostasis → **new 16-Fr silicone Foley** → pass each 3-0 Monocryl through the bladder neck at its matching position, **inside → out** (6-o'clock traction everts mucosa; French-eye needle for 12, 2 and 5 o'clock) → irrigate, test balloon, pass catheter, inflate 15 mL → **Babcock on the anterior bladder holds the neck down while tying — no catheter traction → anterior suture tied first**, then 2, 5, 10, 7, 6 o'clock.
- **No tension;** if present, free the bladder from the peritoneum. Babcock coaptation has **markedly reduced bladder neck contracture**.
- **Closure:** check catheter not caught → irrigate → small suction drain through fascia away from midline, **between (not through) the recti** → running No. 2 nylon, skin clips → catheter taped to thigh.

Postoperative Management

- **Usually uneventful.** The Walsh technique cut **mean stay to about 1.7 days**, with urine leak, transfusion, ileus and re-exploration for bleeding all well under 1%.
- **Mobilise the evening of surgery;** discharge day 1–2. Clear fluids that evening, low-fat diet next day. **Single closed-suction drain** out at discharge or when <50 mL/day off suction.
- **Analgesia:** intravenous patient-controlled opioid (hydromorphone, fentanyl, morphine) or **ketorolac, preferred to spare opioids**; intravenous paracetamol works similarly but is costly. The combination reduces ileus.
- **Ketorolac (non-steroidal anti-inflammatory drug, NSAID) only if** creatinine <1.2 mg/dL, blood loss <1500 mL, **no uncontrolled diabetes, no postoperative haematuria**; given before recovery transfer or next morning.
- **Catheter:** very bothersome, delays return to activity and work → remove as early as safe; the risks are retention and bladder neck contracture. **Never with significant extravasation.**
- **Most anastomoses are watertight on cystography by day 4, yet removal then causes frequent acute retention** — so not recommended.
- **Delayed healing is almost always a perianastomotic haematoma distracting the anastomosis.**
- **Removal at 1 week, three ways:** routine **cystography** (early removal, identifies the few needing longer drainage); **transperineal saline sonocystography** (no contrast cost or radiation); or **no imaging if haematocrit falls <6% from recovery to discharge.**
- **Bladder spasms** usually settle with reassurance; if severe, **oral oxybutynin 5 mg** or **oral diazepam 5–10 mg**.

Concept — *Every step does one of three jobs: secure absolute venous control before touching the apex; keep the right fascia on the specimen; keep traction and heat off the NVB. Apex-first, side-to-side, clip-not-cautery and stapled or sutured DVC control all follow from these.*

COMPLICATIONS

- **ORRP is well tolerated**, with minimal morbidity and **mortality 0.2%**.

Intraoperative Complications

INTRAOPERATIVE COMPLICATIONS: CAUSE AND MANAGEMENT

Complication	Typical setting	Management
Haemorrhage (commonest; usually venous)	Endopelvic fascia incision, puboprostatic ligament division, apical exposure and DVC transection	Packing, exposure, suture or clip ligation; fine 5-0 cardiovascular sutures if needed. Troublesome DVC bleeding → divide the DVC completely over the urethra and oversee the end (single best measure)
Obturator nerve injury	PLND	If divided → reanastomose with fine non-absorbable sutures
Rectal injury	Apical dissection of small prostates , developing the rectum–Denonvilliers plane	Two-layer repair + omental interposition (below); diverting colostomy after previous radiotherapy (salvage RP)
Ureteric injury	Dissecting within the trigone while seeking the bladder–seminal vesicle plane	Ureteric reimplantation

- **Why partial DVC division bleeds more:** exposure needs traction, which pulls partly cut veins open. Fully divided and suture-ligated, the DVC usually stops.
- **Blood loss typically 300–1000 mL; homologous transfusion rarely needed.** No autologous donation, but type and screen for every patient.
- **Rectal repair sequence:** finish prostatectomy → reconstruct bladder neck → perfect haemostasis → small peritoneal opening → **slender, well-vascularised omental pedicle** long enough for the pelvic floor → through an opening in the rectovesical cul-de-sac → assistant **dilates the anal sphincter digitally** to show the injury → **freshen edges, close in two layers** → **omentum fixed over the repair, between rectal closure and anastomosis** → anastomosis → copious antibiotic irrigation → **several days of broad-spectrum aerobic and anaerobic cover**.
- **The omentum prevents rectourethral fistula;** in the reported series there was no wound infection, pelvic abscess or fistula.

Postoperative Complications

- **Life-threatening delayed haemorrhage is rare. Significant bleeding = needing acute transfusion to support blood pressure.** Most are managed expectantly.
- **Explored vs conservative:** similar blood use, but **shorter stay after re-exploration**. Left alone, the haematoma may drain through the anastomosis → **bladder neck contracture and long-term incontinence**; only a minority of explored men have prolonged mild incontinence.
- **Signs of major bleeding in recovery** — hypotension, tachycardia, falling urine output, lower abdominal distension → **early exploration to evacuate the haematoma**, preventing prolonged ileus, anastomotic dehiscence, prolonged catheterisation and contracture. **Immediate exploration after resuscitation** often still allows discharge on day 1–2.

Thromboembolic Events

- **Deep vein thrombosis (DVT) with pulmonary embolism (PE) is the major potential cause of death after RP.**
- **Risk highest with PLND and peaks 14–28 days after surgery** → **stay alert for 4 weeks**.
- **Prevention:** careful positioning to avoid compressing leg veins, **intermittent pneumatic compression, early ambulation**. Some centres add mini-dose heparin or low-molecular-weight heparin (LMWH); the authors do not, as their DVT and PE rates are very low.
- **An informed patient is the best protection:** repeated verbal and written advice to call at once with calf or leg swelling or pain, pleuritic chest pain, haemoptysis, breathlessness, or sudden weakness or fainting.

- **Treatment:** twice-daily subcutaneous LMWH, unmonitored, is at least as effective and safe as adjusted-dose intravenous unfractionated heparin.

Bladder Neck Contracture

- **Historically 0.5–10%**, now much lower.
- **Causes:** poor approximation, urinary extravasation, haematoma distracting the bladder neck.
- **Typically presents at about 3 months** with a poor stream or regressing continence.
- **Treatment ladder:** cystoscopic dilatation → cold-knife incision at 3, 6 and 9 o'clock + short-term intermittent self-catheterisation → recalcitrant: triamcinolone acetonide 200 mg in 5 mL injected after incision → very rarely, reconstruction with an artificial urinary sphincter.

Urinary Incontinence

- **Mechanism:** usually intrinsic sphincter deficiency. Few men over 65 (population registry data) need an incontinence operation within 2 years.
- **Main cause:** sphincter injury during DVC ligation and division. Contributors: a poorly developed sphincter, or a thinner, more collagenous one in older men; smooth muscle damaged by large deep anastomotic sutures or denervated by NVB injury; **contracted or over-wide bladder neck**.
- **Prevention:** preserve the sphincter at the apex; **tension-free** anastomosis; **small, supple bladder neck**; precise **mucosa-to-mucosa** anastomosis.
- **Pre-existing outlet obstruction** (detrusor hypertrophy, poor compliance) slows recovery → **avoid excessive bladder traction** intraoperatively.
- **Until dry:** less fluid; no caffeine or alcohol; **stop α -adrenergic antagonists** taken for hypertension or benign prostatic hyperplasia (BPH). By severity and cause: **imipramine, phenylephrine, anticholinergics or adrenergic agonists**.
- **Continence defined as 0–1 pad per 24 h.**

RECOVERY AFTER ORRP BY AN EXPERIENCED SURGEON (LEPOR CONTINENCE; WALSH POTENCY)

Time after surgery	Continent (0–1 pad/24 h)	Potent (Walsh series)
3 months	80%	38%
6 months	90%	54%
12 months	93%	73%
18 months	—	86%
24 months	97%	—

- **Pattern:** continence returns faster and more completely than potency; both climb steadily, so counsel that recovery is gradual.
- **Trap:** these are single-surgeon, centre-of-excellence figures; another expert centre found **only about a third back to baseline potency at 2 years** after bilateral nerve sparing, despite PDE5 inhibitors.

Erectile Dysfunction

- **Four predictors of recovery:** age <65, preoperative potency, no adult-onset diabetes, bilateral NVB preservation. Age at surgery independently affects recovery.
- **Walsh series** (validated questionnaire, independent third-party review): **86% had unassisted intercourse, with or without sildenafil, at 18 months**; about a third used sildenafil, but only 4% needed it.
- **Most recovery within 2 years; meaningful gains up to 4 years.**
- **High anterior release** → **earlier recovery and return to baseline:** most potent men potent at 12 months, most back to baseline — including those having intercourse more than weekly. Equal results after

unilateral and bilateral release show the gain is from **less traction and more accurate preservation**, not anterior branches; other surgeons report similar gains with early release.

TRIALS OF PDE5 INHIBITOR PENILE REHABILITATION AFTER RP

Trial	Design	Arms	Result
Padma-Nathan	First multicentre randomised placebo-controlled trial	Nightly sildenafil 50 mg vs placebo, 9 months	Better erectile function at 1 year, off drug — first evidence for prophylactic use
REINVENT (Montorsi)	Multi-institutional, randomised, double-blind	Vardenafil on demand vs nightly vs placebo, 9 months	On-demand better than nightly at end of blinded treatment
Sildenafil schedule trial	Randomised, similar design	On-demand vs nightly sildenafil	No difference; trend favouring on-demand

- **Pattern:** rehabilitation helps, but **no trial shows nightly beats on-demand** — daily dosing is not evidence-based.
- **Under-reported effects:** penile shortening and climacturia.

Lower Urinary Tract Symptoms

- ORRP improves lower urinary tract symptoms (LUTS) in men with **moderate-to-severe International Prostate Symptom Score (IPSS)**, who usually gain urinary quality of life **despite incontinence risk**.
- **Benefit lasts up to 15 years** → RP appears to alter the natural history of BPH and LUTS.

Concept — *Most late complications trace to one event, the pelvic haematoma: it distracts the anastomosis, causes contracture and incontinence, and inflames the NVBs. Hence early re-exploration, clips near the bundles and a peritoneal window.*

SUMMARY OF OPEN RADICAL RETROPUBIC PROSTATECTOMY

- **More men are diagnosed younger with curable disease**, and live longer: **a man aged 65–70 has a 50% chance of 15 more years**.
- **ORRP is ideal for men who can be cured and will live long enough to benefit** — who also have the best postoperative quality of life.
- **Robot-assisted RP has spread over the last decade; its main advantage is less blood loss**, and even proponents concede no meaningful overall advantage. Some of the authors still operate only open.

KEY POINTS: ANATOMIC OPEN RADICAL RETROPUBIC PROSTATECTOMY

- **Total removal** = best cure for organ-confined cancer.
- **Goals:** cancer → complications → continence → potency.
- **Defer 6 weeks after biopsy, 12 weeks after TURP.**
- **Fibreoptic headlight:** most useful instrument (surgery beneath the pubis).
- **Grade Group 1–2**, low-volume favourable → **no PLND**.
- **Anterior apical nerve branches** → **high anterior release before DVC division** → earlier potency.
- **Both NVBs usually preserved;** bilateral excision rarely needed.
- **Induration** → **excise bundle; fixed bundle** → **don't preserve;** touch dulled laparoscopically, absent robotically.
- **Transfusion for hypotension** → **explore, evacuate haematoma** (prevents contracture and incontinence).
- **Thromboembolism = major cause of death;** positioning, compression devices, early ambulation.

- **Complete seminal vesicle excision** essential for cancer control.
- **ORRP ideal** for curable men who will live long enough to benefit.